CHAPTER FOUR

RESULT

4.1 Introduction

This chapter presents and analyzes the data obtained from respondents on the perceived effectiveness of non-pharmacological pain management among postoperative patients at State Hospital, Ijebu-Ode, Ogun State. The data were gathered using structured questionnaires administered to selected postoperative patients who met the inclusion criteria. The responses are organized and presented in tables for clarity and better understanding.

The analysis in this chapter is structured according to the study's research objectives and questions. Descriptive statistics such as frequency, mean, and standard deviation are employed to summarize responses on each research variable, while the Chi-square test of independence is used in Section 4.4 to test the stated hypothesis and determine the relationship between the level of knowledge and the perceived effectiveness of non-pharmacological pain management strategies. The discussion of findings is based on the literature reviewed in Chapter Two to provide theoretical and empirical linkage to the results obtained.

4.2 Socio-Demographic Characteristics of Respondents

This section presents the socio-demographic profile of respondents, which includes their age, gender, marital status, educational level, and type of surgery undergone. Understanding these characteristics provides important background information that may influence the respondents' perception and experience of non-pharmacological pain management strategies after surgery.

Table 4.1 reveals that most of the respondents (38.6%) were aged between 18 and 25 years, followed by those aged 26 to 35 years (30.1%), indicating that the majority of postoperative patients in the study were young adults. Female respondents (67.5%) outnumbered their male counterparts (32.5%), which may be attributed to the high number of obstetric and gynecological surgeries such as caesarean sections. Over half of the respondents (55.4%) were single, while 34.9% were married, suggesting that the study captured a relatively youthful population.

Regarding education, most respondents (78.3%) had primary education, implying a modest literacy level among postoperative patients at the facility. The distribution of surgical procedures shows that appendectomy (22.9%) and caesarean section (16.9%) were the most common surgeries, followed by lumpectomy/mastectomy (12.0%) and orthopedic-related procedures (9.6%). This variety indicates a broad representation of postoperative conditions, making the findings on non-pharmacological pain management applicable across different surgical contexts.

Table 4.1: Socio-Demographic Characteristics of Respondents (N = 83)

Variable	Category	Frequency	Percentage (%)
Age (years)	18–25	32	38.6
	26–35	25	30.1
	36–45	8	9.6
	46–60	8	9.6
	60 and above	10	12.0
Gender	Male	27	32.5
	Female	56	67.5
Marital Status	Single	46	55.4
	Married	29	34.9
	Divorced/Separated	3	3.6
	Widowed	5	6.0
Educational Level	No formal education	18	21.7
	Primary education	65	78.3
Type of Surgery	Appendectomy	19	22.9
	Caesarean Section (including variants)	14	16.9
	Hernia Repair	5	6.0
	Laparotomy	4	4.8
	Lumpectomy / Mastectomy	10	12.0
	Dilation and Curettage (D&C)	4	4.8
	Orthopedic Surgeries (Amputation, Tendon repair, Fixation)	8	9.6
	Eye Surgeries (Cataract, Glaucoma)	5	6.0
	Other Surgeries (Thyroidectomy, Prostatectomy, etc.)	14	16.9

4.3 Analysis Based on Research Objectives and Questions

This section presents the analysis of data according to the three research objectives and corresponding research questions of the study. Each subsection highlights one objective and question, using descriptive statistics such as mean and standard deviation to summarize the responses of postoperative patients regarding their knowledge, perceived effectiveness, and factors influencing adherence to non-pharmacological pain management strategies. The results are presented in tables for clarity, followed by brief interpretations to explain the implications of the findings in relation to each objective.

4.3.1 Knowledge of Non-Pharmacological Pain Management Strategies

Table 4.2 shows that respondents generally demonstrated a high level of knowledge about non-pharmacological pain management strategies. The mean values for most items were close to **1.00**, which corresponds to "Yes" on the response scale, indicating positive awareness. The highest level of awareness was observed for the item "relaxation techniques can help reduce postoperative pain" (Mean = 1.08, SD = 0.320), followed by "deep breathing exercise can help manage pain after surgery" (Mean = 1.20, SD = 0.462).

Overall, the results suggest that most postoperative patients at State Hospital, Ijebu-Ode, possess good knowledge of various non-drug pain management methods such as relaxation, deep breathing, cold/heat therapy, and music therapy. The relatively low standard deviations further indicate consistency in their responses, implying that the respondents' knowledge levels are fairly uniform across the sample population.

Table 4.2: Descriptive Statistics on Knowledge of Non-Pharmacological Pain Management Strategies (N = 83)

Statement	N	Mean	Std. Deviation
Have you heard about pain management without drugs before?	83	1.22	0.443
Were you informed about any non-drug pain relief method after surgery?	83	1.33	0.521
Do you understand how these strategies can help relieve pain?	83	1.29	0.530
Are you aware that relaxation techniques can help reduce postoperative pain?	83	1.08	0.320
Do you know that deep breathing exercise can help manage pain after surgery?	83	1.20	0.462
Are you aware that cold/heat therapy can be used for pain relief?	83	1.23	0.477
Do you know that music therapy can help relieve pain after surgery?	83	1.39	0.621

4.3.2 Perceived Effectiveness of Non-Pharmacological Pain Management Strategies

As shown in Table 4.3, respondents generally agreed that non-pharmacological pain management techniques are effective in managing postoperative pain. The mean scores for all items ranged between 3.67 and 3.89, which fall above the midpoint of the 5-point Likert scale, indicating positive perceptions toward these interventions. The highest mean value was recorded for cold or heat therapy (Mean = 3.89, SD = 1.018) and massage therapy (Mean = 3.88, SD = 0.903), suggesting that these two methods were perceived as the most effective for pain relief.

The relatively low standard deviations across items imply that respondents' opinions were consistent. These results indicate that patients at State Hospital, Ijebu-Ode, generally perceive non-pharmacological pain management strategies — particularly relaxation, massage, and cold/heat therapy — as effective adjuncts for postoperative pain relief, supporting the growing recognition of these methods in clinical pain management practice.

Table 4.3: Descriptive Statistics on Perceived Effectiveness of Non-Pharmacological Pain Management

Statement	N	Mean	Std. Deviation
Relaxation techniques (e.g., meditation) are effective in reducing my postoperative pain	83	3.83	1.046
Deep breathing exercise helps me manage my pain after surgery	83	3.75	1.057
Listening to music helps me cope with postoperative pain	83	3.67	1.116
Cold or heat therapy is effective in relieving my surgical pain	82	3.89	1.018
Massage therapy helps reduce my postoperative pain	83	3.88	0.903

4.3.3 Factors Influencing Adherence to Non-Pharmacological Pain Management Strategies

Table 4.4 reveals that respondents generally agreed that the hospital environment, staff support, and availability of necessary materials encourage the use of non-pharmacological pain management strategies. The mean values for most items ranged between 1.12 and 1.27, which corresponds to "Yes" on the response scale, indicating a high level of agreement among participants. The highest mean value (Mean = 1.27, SD = 0.444) was recorded for the statement "I believe non-drug methods can be effective in reducing my pain", showing strong patient confidence in the effectiveness of such methods.

Overall, these findings imply that a supportive environment, adequate materials, and professional guidance from nurses play a significant role in influencing patients' willingness to adopt and adhere to non-pharmacological pain relief strategies. The consistently low standard deviations indicate homogeneity in respondents' opinions, suggesting that most postoperative patients share similar positive perceptions regarding these facilitating factors.

Table 4.4: Descriptive Statistics on Factors Influencing Adherence to Non-Pharmacological Pain Management Strategies

Statement	N	Mean	Std. Deviation
The hospital environment supports relaxation and comfort	83	1.13	0.341
Nurses or health workers are available to assist me with pain relief activities	83	1.12	0.328
The necessary materials for non-drug pain relief (e.g., pillows, cold packs) are provided when needed	83	1.12	0.328
I believe non-drug methods can be effective in reducing my pain	83	1.27	0.444
I am motivated to continue using them if they work for me	83	1.18	0.387

4.4 Test of Hypothesis

This section tests the hypothesis formulated in Chapter One to determine whether there is a significant relationship between the level of knowledge and the perceived effectiveness of non-pharmacological pain management strategies among postoperative patients at State Hospital, Ijebu-Ode, Ogun State. The Chi-square test of independence was used because both variables were measured on categorical scales.

Table 4.5 shows that the Pearson Chi-Square value is 12.950 with a significance level (p-value) of 0.002, which is less than the conventional alpha level of 0.05. This result indicates that there is a statistically significant relationship between respondents' level of knowledge and their perceived effectiveness of non-pharmacological pain management strategies. Hence, the null hypothesis stating that "there is no significant relationship between the level of knowledge and perceived effectiveness of non-pharmacological pain management strategies among postoperative patients" is rejected.

This implies that postoperative patients who possess greater knowledge of non-pharmacological pain management techniques (such as relaxation, deep breathing, and music therapy) tend to perceive these methods as more effective for relieving pain. In essence, improved awareness and understanding of these strategies are likely to enhance patients' belief in their usefulness, supporting the idea that education and health information play a crucial role in promoting non-drug pain control practices.

Table 4.5: Chi-Square Test Showing Relationship Between Knowledge and Perceived Effectiveness of Non-Pharmacological Pain Management

Test	**Value**	**df**	**Asymp. Sig. (2-sided)**
Pearson Chi-Square	**12.950**	**2**	**0.002**
Likelihood Ratio	11.117	2	0.004
Linear-by-Linear Association	11.660	1	0.001
N of Valid Cases	**82**	—	—

CHAPTER FIVE

DISCUSSION, SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Discussion of Findings

The findings of this study revealed that postoperative patients at State Hospital, Ijebu-Ode, possess a high level of knowledge about non-pharmacological pain management strategies. Most respondents demonstrated awareness of techniques such as relaxation, deep breathing exercises, cold/heat therapy, and music therapy. This aligns with the report of Abdullahi et al. (2021), who noted that patient education and health worker guidance significantly improve awareness of non-drug pain management in clinical settings. Similarly, Adeniyi and Olanrewaju (2020) found that postoperative patients who were properly informed about non-pharmacological methods exhibited greater willingness to use them in conjunction with analgesic drugs. These findings confirm that adequate knowledge of pain control methods serves as a foundation for effective postoperative pain management.

The study also revealed that respondents generally perceived non-pharmacological pain management methods as effective, with relaxation techniques, massage, and cold/heat therapy being rated highest. This outcome corroborates the work of Johnson and Afolabi (2019), who reported that patients receiving relaxation and breathing therapy experienced significant reductions in pain intensity after abdominal surgery. Similarly, Odukoya et al. (2018) emphasized that music and massage therapies not only reduce pain but also improve comfort and emotional stability among postoperative patients. These findings collectively demonstrate that non-pharmacological methods play an essential complementary role in pain management and recovery.

Furthermore, the study found that several factors positively influenced adherence to non-pharmacological pain management strategies, including a supportive hospital environment, availability of nurses, and provision of necessary materials. These findings are in agreement with Ogundipe et al. (2020), who noted that hospital environment and staff support were crucial determinants of patient participation in self-care and rehabilitation activities. The results also support the Gate Control Theory of Pain, which posits that non-drug interventions such as relaxation and distraction techniques can modulate pain perception through sensory pathways and psychological mechanisms.

Finally, the Chi-square test showed a statistically significant relationship between knowledge and perceived effectiveness of non-pharmacological pain management strategies ($p = 0.002$). This finding is consistent with the results of Eze and Okonkwo (2018), who found that patients with higher knowledge of pain management approaches were more likely to report positive experiences and higher satisfaction with care. It can therefore be inferred that improving patients' knowledge through education and nurse-led guidance enhances their belief in and adherence to non-drug pain management strategies. Overall, these findings reinforce the need for integrating structured non-pharmacological pain management education into postoperative care protocols in Nigerian hospitals.

5.2 Implications of Study

The findings of this study have several important implications for nursing practice, particularly in the area of postoperative pain management. First, the high level of knowledge and positive perception of non-pharmacological pain management among patients highlights the crucial role of nurses as educators and advocates for holistic care. Nurses are in a strategic position to provide patients with continuous information, demonstrations, and reinforcement on the use of non-drug

pain relief methods such as relaxation, deep breathing, and music therapy. By integrating patient education into routine postoperative care, nurses can empower patients to actively participate in their recovery and reduce reliance on pharmacological interventions.

Secondly, the study underscores the need for nurses to adopt a multidisciplinary approach in managing postoperative pain. Collaboration with physiotherapists, psychologists, and physicians will ensure that patients receive comprehensive pain management that addresses both physical and psychological dimensions. This approach aligns with evidence-based practice, which emphasizes combining pharmacological and non-pharmacological methods for optimal patient outcomes. Nurses should also advocate for institutional policies that promote a supportive environment conducive to non-drug pain management practices, including ensuring privacy, comfort, and access to required materials.

Finally, the significant relationship found between knowledge and perceived effectiveness suggests that continuous professional development for nurses is essential. Training and workshops on evidence-based non-pharmacological interventions should be prioritized to enhance nurses' competence and confidence in applying these strategies. Strengthening nursing practice in this way will not only improve patient satisfaction and comfort but also align with the global nursing goal of delivering safe, patient-centered, and holistic care.

5.3 Limitations of the Study

Like all empirical research, this study was subject to certain limitations that should be acknowledged. First, the study was conducted among postoperative patients in a single healthcare facility — State Hospital, Ijebu-Ode — which may limit the generalizability of the findings to other hospitals or regions with different patient populations, healthcare systems, or resource

availability. Secondly, the study relied on a self-administered questionnaire, and as such, respondents' answers may have been influenced by **self-reporting bias**, misunderstanding of some questions, or the tendency to provide socially desirable responses.

Another limitation was the **sample size** of 83 respondents, which, although adequate for the analysis conducted, may not capture all possible variations in knowledge, perception, and adherence levels among postoperative patients in larger populations. In addition, only descriptive and inferential statistics (mean, standard deviation, and Chi-square) were used; more advanced statistical methods could provide deeper insights into causal relationships among variables. Finally, the cross-sectional nature of the study means that findings represent the situation at one point in time, and not necessarily long-term perceptions or outcomes of non-pharmacological pain management strategies.

5.4 Contribution to Knowledge

This study has contributed significantly to the growing body of knowledge on non-pharmacological pain management, particularly within the Nigerian healthcare context. First, it provides empirical evidence that postoperative patients in a public hospital setting possess a relatively high level of awareness and understanding of non-drug pain relief strategies. This finding adds to existing literature by demonstrating that knowledge dissemination by healthcare professionals—especially nurses—can influence patients' acceptance and utilization of alternative pain management methods.

Secondly, the study established that non-pharmacological interventions such as relaxation, deep breathing, massage, and cold/heat therapy are perceived by patients as effective in relieving

postoperative pain. This reinforces the global emphasis on holistic care and supports the integration of complementary and non-invasive pain control methods into nursing practice in Nigeria.

Furthermore, the research identified specific factors influencing adherence, such as hospital environment, staff support, and availability of materials, thereby providing practical insights for healthcare administrators seeking to improve postoperative care delivery. The significant relationship found between knowledge and perceived effectiveness also contributes to theoretical understanding by supporting the Gate Control Theory of Pain, which explains how cognitive and psychological factors can modulate pain perception. Overall, this study bridges an important knowledge gap and offers a local perspective that can inform future policy, training, and patient-centered care approaches in non-pharmacological pain management.

5.5 Summary of Findings

This study investigated the perceived effectiveness of non-pharmacological pain management strategies among postoperative patients at State Hospital, Ijebu-Ode, Ogun State. The study was guided by three research objectives, three research questions, and one hypothesis. The objectives were to assess the level of knowledge of non-pharmacological pain management strategies among postoperative patients, determine their perceived effectiveness, and identify the factors influencing adherence to such methods. The study employed a descriptive research design with a total of 83 respondents selected through purposive sampling. A structured questionnaire was used for data collection, and the data were analyzed using descriptive statistics such as mean and standard deviation, while a Chi-square test was used for hypothesis testing at a 0.05 level of significance.

Findings from the socio-demographic analysis showed that the majority of the respondents were young adults between 18 and 35 years of age, and females constituted the greater proportion of

participants. Most of the respondents had primary education, and appendectomy and caesarean section were the most common surgical procedures undergone. These demographic characteristics provided a background that helped in understanding patients' responses to non-pharmacological pain management practices.

The first research objective revealed that postoperative patients possessed a high level of knowledge of non-pharmacological pain management strategies. Most of the respondents were aware of relaxation, deep breathing, music, and cold/heat therapy as effective non-drug pain relief methods. This indicates that health workers in the study area may have played a significant role in educating patients about various methods that can be used to manage pain without medication. The high level of awareness suggests that non-pharmacological pain management is becoming increasingly recognized as a complementary approach to pain relief in postoperative care.

The second research objective showed that patients generally perceived non-pharmacological methods as effective in managing postoperative pain. The descriptive statistics revealed that the highest levels of agreement were recorded for cold/heat therapy, massage therapy, and relaxation techniques. This positive perception highlights patients' acceptance of these methods as beneficial in reducing pain and promoting comfort after surgery. It also aligns with findings from previous studies which confirmed that integrating non-drug interventions such as relaxation, breathing, and massage therapy can reduce postoperative discomfort and anxiety.

The third objective identified several factors influencing adherence to non-pharmacological pain management strategies. These include a supportive hospital environment, availability of nurses to assist patients, provision of required materials, and patients' motivation and belief in the effectiveness of such methods. The study further revealed through Chi-square analysis that there was a significant relationship between knowledge and perceived effectiveness of non-

pharmacological pain management ($\chi^2 = 12.950$, $p = 0.002$). This implies that the higher a patient's knowledge of pain management strategies, the more effective they perceive these strategies to be. This relationship underscores the importance of education and professional guidance in enhancing the adoption and effectiveness of non-pharmacological pain management practices.

5.6 Conclusion

Based on the findings, it can be concluded that postoperative patients at State Hospital, Ijebu-Ode, generally possess a high level of knowledge of non-pharmacological pain management strategies. The majority of patients demonstrated awareness of methods such as relaxation, deep breathing exercises, and cold/heat therapy, which they recognized as useful for pain relief and recovery. This suggests that non-drug interventions are gradually being integrated into postoperative pain management in the hospital.

The study also concludes that patients perceive non-pharmacological pain management strategies as effective in relieving postoperative pain. Techniques such as massage, relaxation, and music therapy were viewed as effective in reducing discomfort and anxiety, complementing the use of analgesic medications. Moreover, adherence to these strategies was strongly influenced by environmental support, healthcare personnel assistance, and the availability of essential materials for implementation.

Finally, the study confirmed a statistically significant relationship between knowledge and perceived effectiveness of non-pharmacological pain management. This means that as patients' awareness and understanding of these techniques increase, so does their belief in their effectiveness. Therefore, strengthening patient education and encouraging health professionals to

incorporate non-pharmacological strategies into routine postoperative care will enhance pain management outcomes and improve patient satisfaction.

5.7 Recommendations

Based on the findings and conclusions of this study, the following recommendations are made:

1. **Enhance Patient Education:** Healthcare professionals, particularly nurses, should provide structured health education programs to increase patients' knowledge and understanding of non-pharmacological pain management techniques before and after surgery.

2. **Integrate Non-Pharmacological Methods into Postoperative Care:** Hospitals should formally integrate non-drug pain relief methods such as relaxation therapy, breathing exercises, and music therapy into postoperative nursing care plans to complement pharmacological interventions.

3. **Provide Necessary Materials and Facilities:** Hospital management should ensure the availability of materials and resources (e.g., pillows, warm/cold packs, music systems) that support non-pharmacological pain management practices.

4. **Train Healthcare Personnel:** Continuous professional training should be organized for nurses and other healthcare providers to enhance their skills and confidence in administering and supervising non-pharmacological pain management methods.

5. **Encourage Policy Support:** The Ministry of Health and hospital administrators should develop policies and guidelines that encourage the adoption of evidence-based non-pharmacological interventions as standard practice for postoperative pain management in Nigerian healthcare facilities.

5.8 Suggestions for Further Studies

Based on the scope and limitations of this study, the following areas are suggested for future research:

1. Future studies could examine the long-term impact of non-pharmacological pain management strategies on patients' recovery rate and psychological well-being after surgery.
2. Researchers may conduct comparative studies across multiple hospitals or healthcare settings to determine whether patients' perceptions and adherence differ by institution or region.
3. Further studies should explore the perception and practices of nurses and other healthcare professionals regarding the integration of non-pharmacological pain management techniques in postoperative care.
4. Qualitative research approaches such as interviews or focus group discussions could be used to gain in-depth insights into patients' personal experiences and challenges in using non-drug pain relief methods.
5. Lastly, studies could assess the cost-effectiveness and patient satisfaction outcomes of combining pharmacological and non-pharmacological interventions for pain management in Nigerian hospitals.

APPENDICES

APPENDIX 1: QUESTIONNAIRE

Dear Respondent

I am an undergraduate of the above-named school and currently conducting a study on **perceived effectiveness of non-pharmacological pain management among postoperative patients at State Hospital Ijebu Ode, Ogun State**. I would be thankful if you could spare a few minutes of your fill this questionnaire. There will be no incentives for participating in this study. In addition to this, you will not be affected in any way if choose to participate and your answers will be anonymous and treated with utmost confidentiality. This questionnaire is to assess the perceived effectiveness of non-pharmacological pain management strategies on postoperative patients.

SECTION A: Socio- Demographic Characteristics

1. Age: _____ 18-25 yrs [ ] 25-35 yrs [ ] 36-45 yrs [ ] 46-60 yrs [ ] 60 and above [ ]

2. Gender: [ ] Male [ ] Female

3. Marital Status: [ ] Single [ ] Married [ ] Divorced/Separated [ ] Widowed

4. Educational Level: [ ] No formal education [ ] Primary [ ] Secondary [ ] Tertiary

5. Type of Surgery Undergone: _____

6. Duration Since Surgery: [ ] Less than 1 week [ ] 1-2 weeks [ ] 2-4 weeks [ ] More than 4 weeks

Section B: Knowledge of non-pharmacological pain management strategies

Statement	Yes	No	Not Sure

8. Have you heard about pain management without drugs before?			
10. Were you informed about any non-drug pain relief method after your surgery?			
11. Do you understand how this strategies can help relieve pain?			
12. Are you aware that relaxation techniques can help reduce post operative pain?			
13. Do you know that deep breathing exercise can help manage pain after surgery?			
14. Are you aware that cold/heat therapy can be used for pain relief?			
15. Do you know that music therapy can help relieve pain after surgery			

Section C: Perceived Effectiveness of Non-Pharmacological Pain Management Strategies.

(Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree)

Statement	**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**
16. Relaxation techniques (e.g meditation) are effective in reducing my postoperative pain.					
17. Deep breathing exercise help me manage my pain after surgery.					
18. Listening to music help me cope with post operative pain.					

19. Cold or heat therapy is effective in relieving my surgical pain					
20. Massage therapy helps reduce my post operative pain					

Section D: Factors Influencing Adherence to Non-Pharmacological Pain Management Strategies

STATEMENT	YES	NO
21. The hospital environment support relaxation and comfort.		
22. Nurses or health workers are available to assist me with pain relief activities.		
23. The necessary material for non-drug pain relief (e.g pillows, cold packs) are provided when needed.		
24. I believe non drug methods can be effective in reducing pain		
25. I am motivated to continue using them if they work for me.		

Thank you for your participation!

APPENDIX 2: AGE RANGE

Age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-25 years	32	38.6	38.6	38.6
	26-35 years	25	30.1	30.1	68.7
	36-45 years	8	9.6	9.6	78.3
	46-60 years	8	9.6	9.6	88.0
	60 and above	10	12.0	12.0	100.0
	Total	83	100.0	100.0	

APPENDIX 3: GENDER

Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	27	32.5	32.5	32.5
	Female	56	67.5	67.5	100.0
	Total	83	100.0	100.0	

APPENDIX 4: MARITAL STATUS

Marital Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Single	46	55.4	55.4	55.4
	Married	29	34.9	34.9	90.4
	Divorced/Separated	3	3.6	3.6	94.0
	Widowed	5	6.0	6.0	100.0
	Total	83	100.0	100.0	

APPENDIX 5: EDUCATIONAL LEVEL

Educational Level

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal education	18	21.7	21.7	21.7
	Primary	65	78.3	78.3	100.0
	Total	83	100.0	100.0	

APPENDIX 6: TYPE OF SURGERY UNDERGONE

Type of Surgery Undergone

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Amputation	3	3.6	3.6	3.6
	Appendectomy	18	21.7	21.7	25.3

Appendectomy and childbirth	1	1.2	1.2	26.5
Bowel resection	1	1.2	1.2	27.7
Caesarean section	4	4.8	4.8	32.5
Caesarean Section	1	1.2	1.2	33.7
cataract surgery	1	1.2	1.2	34.9
Cataract surgery	3	3.6	3.6	38.6
ceasaren section	1	1.2	1.2	39.8
Ceasaren section	7	8.4	8.4	48.2
Ceaserian section	1	1.2	1.2	49.4
Dilation and curettage	3	3.6	3.6	53.0
Dilation and Curettage	1	1.2	1.2	54.2
External fixation of fracture	1	1.2	1.2	55.4
External Fixation of fracture	1	1.2	1.2	56.6
External fixation of fractures	1	1.2	1.2	57.8
External Fracture Fixation	1	1.2	1.2	59.0
Glaucoma	1	1.2	1.2	60.2
Glaucoma surgery	1	1.2	1.2	61.4
hernia repair	1	1.2	1.2	62.7
Hernia repair	3	3.6	3.6	66.3
Hernia Repair	1	1.2	1.2	67.5
hysterectomy	1	1.2	1.2	68.7
laparotomy	1	1.2	1.2	69.9
Laparotomy	3	3.6	3.6	73.5
lumpectomy	2	2.4	2.4	75.9
Lumpectomy	6	7.2	7.2	83.1
Mascctomy	1	1.2	1.2	84.3
Masectomy	1	1.2	1.2	85.5
Myomectomy	3	3.6	3.6	89.2
Prostatectomy	1	1.2	1.2	90.4
Spinal surgery	1	1.2	1.2	91.6
Tendon repair	3	3.6	3.6	95.2
Thoracentesis	1	1.2	1.2	96.4
Thyroidectomy	2	2.4	2.4	98.8
Tonsillectomy	1	1.2	1.2	100.0
Total	83	100.0	100.0	

APPENDIX 7: Descriptive Stat for Knowledge of Non-Pharmacological Pain Management Strategies

Descriptive Statistics

	N	Mean	Std. Deviation
Have you heard about pain management without drugs before?	83	1.22	.443
Were you informed about any non- drug pain relief method after surgery?	83	1.33	.521
Do you understand how these strategies can help relieve pain?	83	1.29	.530
Are you aware that relaxation techniques can help reduce post operative pain?	83	1.08	.320
Do you know that deep breathing exercise can help manage pain after surgery?	83	1.20	.462
Are you aware that cold/heat therapy can be used for pain relief?	83	1.23	.477
Do you know that music therapy can help relieve pain after surgery?	83	1.39	.621
Valid N (listwise)	83		

APPENDIX 8: Descriptive Stat for Perceived Effectiveness of Non-Pharmacological Pain Management Strategies

Descriptive Statistics

	N	Mean	Std. Deviation
Relaxation techniques (e.g meditation) are effective in reducing my Postoperative pain	83	3.83	1.046
Deep breathing exercise help me manage my pain after surgery?	83	3.75	1.057
Listening to music help me cope with post operative pain	83	3.67	1.116
Cold or heat therapy is effective in relieving my surgical pain	82	3.89	1.018
Massage therapy helps reduce my Postoperative pain	83	3.88	.903
Valid N (listwise)	82		

APPENDIX 9: Descriptive Stat for Factors Influencing Adherence to Non-Pharmacological Pain Management Strategies

Descriptive Statistics

	N	Mean	Std. Deviation
The hospital environment support relaxation and comfort	83	1.13	.341
Nurses or health workers are available to assist me with pain relief activities	83	1.12	.328
The necessary material for non-drug pain relief (e.g pillows, cold packs are provided when needed	83	1.12	.328
I believe non drug methods can be effective in reducing my pain	83	1.27	.444
I am motivated to continue using them if they work for me	83	1.18	.387
Valid N (listwise)	83		

APPENDIX 10: Chi-Square Test Showing Relationship Between Knowledge and Perceived Effectiveness of Non-Pharmacological Pain Management

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Knowledge Categories * Effectiveness Category	82	98.8%	1	1.2%	83	100.0%

Knowledge Categories * Effectiveness Category Crosstabulation

	Effectiveness Category			Total
	1.00	2.00	3.00	

Knowledge Categories	1.00	Count	4	23	41	68
		Expected Count	7.5	24.0	36.5	68.0
	2.00	Count	5	6	3	14
		Expected Count	1.5	5.0	7.5	14.0
Total		Count	9	29	44	82
		Expected Count	9.0	29.0	44.0	82.0

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	12.950[a]	2	.002
Likelihood Ratio	11.117	2	.004
Linear-by-Linear Association	11.660	1	.001
N of Valid Cases	82		

a. 2 cells (33.3%) have expected count less than 5. The minimum expected count is 1.54.